7.2 Separable First-Order Differential Equations; Uninhibited and Inhibited Growth and Decay Models

Name _____ Period ____ Date _____

AP Calculus 7.2 · Student sheet

I can…

- separate the variables and solve a separable differential equation.
- solve growth and decay problems with A = A_0e^(kt).
- use Newton's Law of Cooling.

Vocabulary · fill in the missing word

Separable equation — an equation that can be _____ as g(y) dy = f(x) dx

Uninhibited growth or decay — $\frac{dA}{dt} = kA$, with _____ $A = A_0e^{kt}$

Half-life — the time for a _____ amount to fall to half

Newton's Law of Cooling — $\frac{du}{dt} = k(u - T)$, where T is the _____ temperature

Part 1 · Penny doubling

A penny that doubles every day.

Day	0	1	2	3	5	10
Cents						

Part 2 · Notes and examples (fill in each note, then work the example)

Separate: put every y with _____ and every x with dx.

Integrate both sides and include one _____ of integration.

Example 1

Solve $\frac{dy}{dx} = 2xy$ with y(0) = 3.

Example 2

Solve $\frac{dy}{dx} = \frac{x}{y}$ with y(0) = 4, y > 0.

Uninhibited growth: dA/dt = kA has solution A = _____.

Example 3

Bacteria double every 5 hours. There are 200 now. How many after 15 hours?

Example 4

A substance has a half-life of 10 years. Find k in $A = A_0 e^{kt}$.

Cooling: u(t) = T + (u_0 − T)e^(kt), where T is the _____ temperature.

Example 5

Coffee at 90° sits in a 20° room. The difference halves every 10 minutes. Find the temperature after 20 minutes.

Part 3 · Partner practice

Partner 1

Solve $\frac{dy}{dx} = 4xy$ with y(0) = 2.

Partner 2

(AP® Practice) The solution of $\frac{dy}{dx} = \frac{x}{y}$ with y(0) = 1 is

(A) $y = \sqrt{x^2 + 1}$ (B) y = x + 1

(C) $y = x^2 + 1$ (D) $y = -\sqrt{x^2 + 1}$